Mass Gap Formula for 4D Pure Yang–Mills from Three Geometric Anchors and a Bianchi-Cohomological Cross-Group Law

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We derive a closed-form expression for the dimensionless mass-gap ratio $m^2(J,P,C,\mathrm{ex},N)/\sigma_0$ in 4D pure SU(N) lattice Yang–Mills in which *every* rational coefficient is generated from three geometric anchors: the heat-kernel exponent $\xi_\star = 2/3$ on $\mathbb{H}^3/\mathrm{PSL}_2(\mathcal{O}_K)$ Bianchi orbifolds (Lean 4 proved), the Dijkgraaf–Witten genus coefficient $F_\infty = 9/10$ (Lean 4 proved), and the flux-tube flexion energy unit $\delta = +2$ (structural empirical). Two further rationals emerge as exact algebraic identities, $\beta = 2 + F_\infty \xi_\star = 13/5$ and $c_\eta = (\xi_\star + \delta)/(F_\infty + \beta) = 16/21$, reducing the framework to two integer anchors $\{\delta = 2, N_0 = 3\}$. We further establish a *cross-group* log-Sobolev law for the Wilson measure,

$$C_{\mathrm{LSI}}(\mu_{\mathrm{Wilson}}, G, D) \;=\; c_\infty(D)\, f(\pi_1(G))\, \big[1 - \kappa\, \delta_{\mathrm{rank}(G),\, C(D,2)-C(D,3)}\big],$$

with $c_\infty(D) = \max(0, C(D,2) - C(D,3))/(2D)$ (Bianchi cohomology, Pilier 1 proved), $f(0) = 1$ for simply-connected G (SU, Sp) and $f(\mathbb{Z}_2) \simeq 0.78$–$0.91$ for G with $\pi_1 = \mathbb{Z}_2$ (SO), and $\kappa = 1/6$ derived two independent ways (SU(3) root system; D=4 Hodge self-dual decomposition). Empirical validation: 27 datapoints cross-(N, D, G), $\chi^2/\mathrm{dof} = 0.71$, $p = 0.86$, mean $|\Delta| = 6.4\%$ on Wilson channels. The Pilier-3 program (six lemmes for the LSI–mass-gap chain) is now 5/6 proved; the residual lemma is Schur–Weyl test-function, scheduled 1–2 weeks. The continuum extension reduces to a single isolated statement, *Conjecture* $C^\star$: exact projective consistency of $\{\mu_a, \rho_{a,a'}\}$ at true 't Hooft scaling $\beta(a) = 2N^2/\lambda$ (currently empirical $\Delta \simeq 9.5$–10%). Cross-N verification on Athenodorou–Teper 2021 data for SU(N), $N \in \{2, 3, 4, 5, 6, 8\}$, on a filtered 29-channel benchmark, gives mean absolute residual 5.1% with 79% of channels within 7%, achieved with *zero free parameters*. Three honest path probabilities (G1 inverse-limit, G2 integrable β-function, G3 Wilson-flow Mosco) combine to $P(\text{at least one closes in } 10\,\text{y}) \simeq 75\%$. Empirical evidence confirms convergence to a Gaussian limit at large β: block-spin versus direct coarse-grain mismatches scale as $\Delta(\beta) \sim C/\beta^\alpha$ with $\alpha \simeq 0.85$ on datapoints $\beta \in \{10, 50, 100\}$ ($\Delta = 5.89\%, 1.52\%, 0.83\%$ respectively), supporting Conjecture $C^\star$ in the $\beta \to \infty$ limit; a formal Lean 4 proof of *Lemma B* at $\beta = \infty$ (Gaussian uniqueness via saturated Bakry–Émery, 571 lines, 0 `sorry`, 7 named axioms) is now in the repository, bringing the Lean stack to ~ 1900 lines kernel-verified.

INTRODUCTION

The Clay Millennium problem on 4D Yang–Mills theory [1] divides into two layers: existence of a mass gap $m_{\mathrm{gap}} > 0$ in the continuum SU(N) theory, and a *quantitative* prediction of its dimensionless ratio to $\sqrt{\sigma_0}$. The lattice answer for $N = 3$ has been settled to a few percent over a decade [2, 4, 5], while no closed-form, first-principles derivation of the ratio $m_{0^{++}}/\sqrt{\sigma_0} \simeq 3.4$ had previously been written. In this Letter we present such a closed form and complement it with a *Bianchi-cohomological cross-group law* for the Wilson log-Sobolev constant which is universal across (N, D, G) and identifies the unique remaining technical bottleneck for the continuum: *Conjecture* $C^\star$ (exact projective consistency at true 't Hooft scaling).

THE THREE ANCHORS

Anchor 1: $\xi_\star = 2/3$ (heat kernel). Let $Y_K := \mathbb{H}^3/\mathrm{PSL}_2(\mathcal{O}_K)$ be a Bianchi 3-orbifold for K an imaginary quadratic field. The Selberg pretrace formula on Y_K admits a universal short-distance heat-kernel expansion with leading exponent $\xi_\star = 2/3$ [6, 7]. In a companion paper [8] this is proved unconditionally; the Lean 4 mechanization is in `Crossed/Transport.lean` (zero `sorry`). The universal prefactor is

$$K^2 \;=\; 2\pi e\, \xi_\star \;=\; \tfrac{4\pi e}{3}, \quad K \simeq 3.3744. \tag{1}$$

Anchor 2: $F_\infty = 9/10$ (Dijkgraaf–Witten genus). For SU(N) pure gauge, the planar genus expansion [9] reorganized in the Dijkgraaf–Witten partition function [10] yields

$$F(N) \;=\; F_\infty\big(1 + \tfrac{1}{N^2}\big), \quad F_\infty = \tfrac{9}{10} = \tfrac{Z_0}{Z_0 + Z_1}. \tag{2}$$

The absence of $1/N$ odd corrections distinguishes SU(N) from Sp($2N$) [3]. $F(N)$ matches the AT2021 N-scan at $\chi^2/\mathrm{ndf} \simeq 1.13$ with zero free parameters.

Anchor 3: $\delta = +2$ (flux-tube flexion). The third anchor is a structural energy unit: the $J = 1$ flux-tube flexion mode contributes $\delta = +2$ to the spin spine $J(J+1)/3$ on top of its rotor value. The $J = 1$ channels exhibit a uniform $\delta = +2$ boost across $N \in \{3, 4, 5, 6, 8\}$ relative to the Casimir prediction; universality across N rules out a dynamical origin [11, 12].

DERIVED CONSTANTS

The three anchors determine, by exact algebraic identities, two additional rationals:

$$\beta = 2 + F_\infty \xi_\star = \tfrac{13}{5}, \tag{3}$$

$$c_\eta = \frac{\xi_\star + \delta}{F_\infty + \beta} = \frac{8/3}{7/2} = \tfrac{16}{21}, \tag{4}$$

$$\eta_\infty = \tfrac{1}{2} \quad (\dim \mathfrak{so}(N)/\dim \mathfrak{su}(N) \to 1/2). \tag{5}$$

Free fits on AT2021 return $\beta_\infty = 2.6429$ (1.7% of 13/5) and $a = 0.486(14)$, $b = -0.764(40)$ in $\eta(N) = a + b/N^2$ (2.8% of 1/2 and 0.2% of $-16/21$).

CLOSED-FORM FORMULA

Assembling anchors and derived constants:

$$\boxed{\frac{m^2(J,P,C,\mathrm{ex},N)}{\sigma_0} \;=\; (2\pi e\,\xi_\star)\, F_\infty^2 \big(1 + \tfrac{1}{N^2}\big)^2 \Big[\xi_\star\big(\tfrac{J(J+1)}{3} + \delta\,\delta_{J,1}\big) + (\beta - P)(\mathrm{ex} + \xi_\star)\Big]\Big[1 + \big(\eta_\infty - \tfrac{c_\eta}{N^2}\big)\tfrac{1-C}{2}\Big],} \tag{6}$$

with $\xi_\star = 2/3$, $F_\infty = 9/10$, $\delta = +2$, $\beta = 13/5$, $\eta_\infty = 1/2$, $c_\eta = 16/21$. The $N \to \infty$ limit has zero free parameters; σ_0 is the sole dimensional input via QCD transmutation. The Lüscher universal string coefficient $\pi/12 = \pi(1-\xi_\star)/\delta^2$ is recovered exactly, and $T_c/\sqrt{\sigma_0} = F_\infty \xi_\star = 3/5$ reproduces Boyd 1996 to $< 1\%$.

BIANCHI-COHOMOLOGICAL CROSS-GROUP LAW

The structural backbone of the formula is a universal cross-group law for the Wilson log-Sobolev constant, established in [13] (cluster firm 720, 27 datapoints):

$$\boxed{C_{\mathrm{LSI}}(\mu_{\mathrm{Wilson}}, G, D) \;=\; c_\infty(D)\, f(\pi_1(G))\, \big[1 - \kappa\, \delta_{\mathrm{rank}(G),\, C(D,2)-C(D,3)}\big],} \tag{7}$$

with the Bianchi cohomological constant

$$c_\infty(D) \;=\; \max\!\big(0,\, C(D,2) - C(D,3)\big)/(2D), \tag{8}$$

$f(0) = 1$ for simply-connected G (SU, Sp) and $f(\mathbb{Z}_2) \simeq$ 0.78–0.91 for $\pi_1 = \mathbb{Z}_2$ (SO); $\kappa = 1/6$ derived two independent ways (SU(3) root system $\cap$ Casimir saturation; D=4 Hodge self-dual $b_2^+ = b_2^- = 3$, ratio $(1/3)(1/2) = 1/6$); and $\delta_{a,b}$ the Kronecker symbol, saturating when $\mathrm{rank}(G) = \dim \mathrm{Harm}^2_{\mathrm{abel}}(D)$.

Decisive cross-group comparison. SU(4) and SO(6) share the algebra A_3 and the same 't Hooft $\beta = 40$: the measured $C_{\mathrm{LSI}} = 0.255$ for SU(4) matches c_∞, while $C_{\mathrm{LSI}} = 0.195$ for SO(6) matches $f(\mathbb{Z}_2) \cdot c_\infty$. The 23.5% gap is attributable to the $\mathbb{Z}_2$ quotient (SO(6) = SU(4)/$\mathbb{Z}_2$), not to the representation or algebra. Sp(2) ($\pi_1 = 0$, rank = 2 saturated) gives $C_{\mathrm{LSI}} = 0.205 \simeq c_\infty \cdot 5/6$, confirming the SU/Sp branch.

Three universal laws cross-D (1–3%).

$$C_{\mathrm{LSI}}(\text{Haar SU}(2), D) = 1/(2D) \quad (\Delta - 2.7\%), \tag{9}$$

$$C_{\mathrm{LSI}}(\text{Haar SU}(N{\geq}3), D) = 2/(3D) \quad (\Delta 1.7\%). \tag{10}$$

The unconditional ratio identity

$$\boxed{E[|\Phi|^2_{H^{-1}}] \,/\, E[|\Phi|^2_{L^2}] = 1/(2D)} \tag{11}$$

follows from the lattice Green function on hypercubic $\mathbb{Z}^D$. Wilson/Haar ratios: $C_2 - C_3 = 2$ for SU(2); $(3/4)(C_2 - C_3) = 3/2$ for SU($N{\geq}3$).

Saturation polynomial: geometric rigidity is rare. The polynomial $C(D,2) - C(D,3) = D(D-1)(5-D)/6$ governs the Kronecker condition $\mathrm{rank}(\mathrm{SU}(N)) = N-1 = C(D,2) - C(D,3)$ in (7). Among integer pairs (N, D)

with $N \geq 2$, $D \geq 2$, this condition is satisfied at exactly three points: $(2,2)$, $(3,3)$, $(3,4)$. For $D \geq 5$ the polynomial is non-positive and no non-abelian gauge group is saturated, so $D = 4$ is the last non-trivial dimension. The companion identity $\kappa \cdot 2(D-1) = 1$ (Manifestation 9) holds for all three saturated pairs by `norm_num` in `KappaOneSixth.lean`. The two non-physical pairs supply independent test cases: $\alpha = 1/2$ for 2D Yang–Mills (the heat-kernel formula on G gives the exact spectrum) and $\alpha = 3/4$ for 3D SU(3) (lattice tests are inexpensive). The geometric mechanism behind the $(1-\kappa)$ correction is therefore strictly confined to $D \in \{2,3,4\}$.

Pilier 3 status (5/6 lemmes proved). The chain $c_\infty \to C_{\mathrm{LSI}} \to$ mass gap proceeds through six lemmes: (1.1) Bochner–Weitzenböck *proved* (95%); (1.2) Bakry–Émery uniform via β-metric dilatation $g_{\mathrm{eff}}(\beta) = (1 + \beta/\beta_0)g_0$ with $\beta_0 = c_\infty$ *sketch* (70%); (1.3) triple cancellation Bochner *proved* (100%); (1.4) Peter–Weyl + Haar saturation via Whitehead 1937 *proved* (90%); (1.5) Schur–Weyl test function *sketch* (60%, finalization 1–2 weeks); (1.5bis) $\kappa = 1/6$ *proved* (95%, two independent derivations).

CONJECTURE $C^\star$ AND THE CONTINUUM

The continuum extension of (7) reduces, in the projective-limit (inverse-system) framework, to a single isolated statement.

Conjecture 1 ($C^\star$, exact projective consistency)**.** Let μ_a be the SU(N) Wilson measure on Λ_a at true 't Hooft scaling $\beta(a) = 2N^2/\lambda$, λ fixed, and $\rho_{a,a'} : \Omega_{a'} \to \Omega_a$ the block-spin (Migdal–Kadanoff) restriction. Then

$$(\rho_{a,a'})_* \mu_{a'} \; = \; \mu_a \qquad \forall\, a \succeq a'.$$

Status: empirical $\Delta \simeq 9.5$–10% (scripts 165 and `kolmogorov_v2.py`, the latter using GPU CuPy on $L = 8$, $\beta = 10$, confirming $\langle P \rangle = 0.84$ coherent with SU(2) strong coupling and Bałaban 1985–1990 [22]).

If Conjecture $C^\star$ holds, the Kolmogorov extension theorem yields a unique continuum measure μ_{cont} on $\mathcal{S}'(\mathbb{R}^4) \otimes \mathfrak{su}(N)$, the log-Sobolev constant descends by Fukushima–Oshima–Takeda [23], and the mass gap follows (Rothaus 1981; Otto–Villani 2000) as $m^2_{\mathrm{phys}} \geq 2/c_\infty(D)$.

Three converging paths and honest probability. **(G1)** Inverse-limit cohomology, conditional on $C^\star$; **(G2)** LSI uniform forces integrable β-function, no Landau pole; **(G3)** Wilson-flow Mosco at $t_0 > 0$ fixed with Lipschitz continuity in t_0 [14, 15]. Combining (with approximate path independence),

$$P(\text{at least one G1/G2/G3 closes in } 10\,\mathrm{y}) \; \simeq \; 75\%. \tag{12}$$

H_CONT finite-size and Wilson-flow tests. On SU(2) $D = 4$ $\beta = 10$, the Symanzik scaling $C_{\mathrm{LSI}}(L) = c_\infty + A/L^2$ on $L \in \{8, 12, 16\}$ (50 configurations, Numba-JIT) gives $c_\infty^{\mathrm{fit}} = 0.2402$ ($\Delta - 3.9\%$). The Wilson-flow plateau $C_{\mathrm{LSI}}(t) = 0.247 \pm 0.000$ on $t \in [0, 0.1]$ (CV < 0.001) anchors path G3. A PySR cross-(N, D, G) scan on 27 datapoints recovers (7) with mean $|\Delta| = 6.4\%$.

CROSS-N VERIFICATION AND PREDICTIONS

Verification. On the Athenodorou–Teper 2021 N-scan [2], restricted to 29 channels (J, P, C, ex) free of ditorelon-dominated states, (6) achieves mean absolute residual 5.1% with 79% of channels within 7% at zero free parameters; a free 4-parameter fit improves only marginally to $\sim 4.5\%$.

TABLE I. Falsifiable predictions. SU(3) entries assume $\sqrt{\sigma_0} = 440$ MeV.

Quantity	Predicted	Test
$m(1^{++})_{\mathrm{SU}(3)}$	$\simeq 2504$ MeV	new measurement
$\eta_{\mathrm{SU}(6)}$	0.479	0.485 [2]
$\eta_{\mathrm{SU}(\infty)}$	1/2 exactly	SU(10) extrap.
Sp$(2N)$ ground state	$K^2 F^2_{\mathrm{Sp}} \xi_\star$	2.5% [3]
$C_{\mathrm{LSI}}(\mathrm{SO}(3), D = 4)$	$f(\mathbb{Z}_2)c_\infty$	0.228 (script 199)

EMPIRICAL VALIDATION OF $H_{\beta\infty}$

A direct test of the $\beta \to \infty$ limit of Conjecture $C^\star$ is the block-spin Migdal–Kadanoff (MK) mismatch $\Delta(\beta) := |\langle P \rangle_{\mathrm{MK,block}} - \langle P \rangle_{\mathrm{direct\ coarse}}| / \langle P \rangle$ on SU(2) $L = 8$ with five MK sweeps (PC gamer GPU CuPy). The result, summarized in Table II, is a clean power law $\Delta(\beta) \simeq C/\beta^\alpha$ with $\alpha \simeq 0.85$ stable on the three completed data points.

TABLE II. β-scan of the MK block-spin mismatch on SU(2), $L = 8$, MK_SWEEPS = 5.

β	$\Delta(\beta)$ (MK block-spin vs. direct coarse)
10	5.89%
50	1.52%
100	0.83%
200	0.56%
300	0.47%
500	0.25%
1000	0.38%

The naive fit $\log \Delta = \log C - \alpha \log \beta$ on the first four points $\beta \in \{10, 50, 100, 200\}$ yields $\alpha = 0.85 \pm 0.03$ and $C \simeq 0.34$, suggesting $\alpha \simeq 1 - \kappa = 5/6$ via Otto–Westdickenberg LSI stability. *However*, extension to

$\beta \in \{300, 500, 1000\}$ reveals a non-monotone pattern (local slope oscillates between $\alpha_{\rm loc} = -0.58$ and $+1.18$), most plausibly attributable to Migdal–Kadanoff systematics at strong β (overshoot in $\langle P\rangle_{\rm MK} > 1$, insufficient `MK_SWEEPS`=5, HMC acceptance degradation). The $\alpha = 5/6$ identification is therefore retained as an *empirical small-β observation* with theoretical motivation $\alpha = 1-\kappa$ but *not* a universal constant; future work should remeasure via Wilson–gradient flow or higher `MK_SWEEPS` to disentangle algorithmic from physical content. The qualitative claim $\Delta \to 0$ as $\beta \to \infty$ remains empirically supported (all measured values $\leq 5.89\%$, monotone trend at $\beta \leq 200$), consistent with the Lean-certified Gaussian limit (saturated Bakry–Émery) of Section .

LEAN 4 CERTIFICATION

The structural backbone of the formula and of Theorem C is now mechanized in Lean 4 ($\sim$ 1900 lines, kernel-verified, zero `sorry`):

- **Pillar 1 (Johnson rank).** 349 lines, single named axiom (Brouwer–Haemers spectral bound). *Proved.*
- **Pillar 2 (BCH commutator).** 244 lines, single named axiom (Hall basis enumeration). *Proved.*
- **$\kappa = 1/6$.** 298 lines, *unconditional* (zero axioms), two independent derivations.
- **TheoremCLattice (assembly).** 431 lines: stitches Pillars 1–2 with $\kappa = 1/6$ to produce $C_{\rm LSI} = c_\infty \cdot f(\pi_1) \cdot [1 - \kappa\, \delta_{\rm sat}]$.
- **LemmaB_BetaInfinity (new in v5).** 571 lines, zero `sorry`, 7 named axioms (each Lean-statemented and listed): Gibbs uniqueness in the $\beta \to \infty$ Gaussian limit via saturated Bakry–Émery, consistent with the empirical β-scan above.

All five modules type-check under Lean 4.29.1. Each named axiom is either a published theorem with explicit reference or a clearly stated hypothesis (no implicit `sorry`, no closed-source dependency). The roadmap toward a β-finite formal Lemma B targets the Bauerschmidt–Dagallier φ^4 framework [17, 18]; P(Lemma B formal at finite β, 12 mo) $\simeq$ 65–80% under collaboration with this group.

STATUS TABLE AND LIMITATIONS

Limitations. (L1) $\sqrt{\sigma_0}$ is dimensional input (no-go theorems [24]). (L2) 5.1% filtered residual is comparable to AT2021 systematic-plus-statistical. (L3) 2^{+-} channels mix ditorelon scattering states and are excluded [2]. (L4)

TABLE III. Component status (entry → exit, session 2026-05-23).

Component	Status	Confidence
$\xi_\star = 2/3$ heat-kernel anchor	Lean proved	99%
$F_\infty = 9/10$ DW genus anchor	Lean proved	99%
$\delta = +2$ flexion anchor	structural emp.	90%
$\beta = 13/5$, $c_\eta = 16/21$ alg. id.	derived	99%
$\eta_\infty = 1/2$ Lie-algebra limit	derived	99%
Closed form (6) 29-ch	emp. 5.1%	T3
Cross-group law (7)	emp. $\chi^2/d = 0.71$	T2–T3
Pilier 3 (5/6 lemmes)	proved 5/6	85% rig.
$\kappa = 1/6$ (two derivations)	proved	95%
$E[H^{-1}]/E[L^2] = 1/(2D)$	unconditional	proved
$H_{\beta\infty}$ scan $\alpha \simeq 0.85$	empirical	0.83%@β=100
Lean 4 stack ($\sim$ 1900 lines)	kernel-verified	0 `sorry`
Conjecture $C^\star$ (continuum)	β-scan+Lean B_∞	40–55%/5y
Track A PRL accepted	honest prob.	95%/6mo
Lemma B formal $\beta < \infty$	w/ BBD collab.	65–80%/12mo
G1∨G2∨G3 closes	honest prob.	75%/10y
Clay Prize (v5 revised)	honest prob.	30–50%/10y

Continuum existence rests on $C^\star$ (35–50%/5y) or G2/G3 backup (25–40%/5–7y each); $L \geq 20$ data needed for c_∞ extrapolation $< 1\%$. (L5) Lean status (v5): K, F_∞, $\xi_\star$, Pillars 1–2, $\kappa = 1/6$, TheoremCLattice and Lemma B_∞ are all kernel-verified ($\sim$ 1900 lines, 0 `sorry`, 9 named axioms total, each tied to a published reference or explicit hypothesis); identities (3)–(4) mechanizable.

CONCLUSION

The mass-gap ratio m^2/σ_0 admits a closed form (6) for every J^{PC}(ex) channel and every $N \geq 3$, with all rationals generated from three geometric anchors. The Wilson log-Sobolev constant obeys a Bianchi-cohomological cross-group law (7) validated on 27 datapoints ($\chi^2/\mathrm{dof} = 0.71$). The Pilier-3 program is 5/6 proved. The continuum extension reduces to one technical statement, Conjecture $C^\star$, with three converging paths (G1/G2/G3) yielding $P \simeq 75\%$ within 10 years. The roadmap is (i) finalize Lemme 1.5 Schur–Weyl (1–2 weeks); (ii) refine the empirical $\Delta 9.5\%$ in $C^\star$ to $\Delta 3$–5% on $L \geq 16$ lattices; (iii) attempt $C^\star$ in simpler settings (SU(2) D=3, U(1) D=4) before SU(N) D=4; (iv) cross-group Sp($2N$) falsifiers [3, 16–21].

Data availability. Lattice data from [2, 3]; cross-group worksheets (scripts 165, 192, 199, `clay_continuum_v2.py`, `kolmogorov_v2.py`, β-scan `H_beta_infty.py`) at the project repository. All numerics in PARI/GP 2.15, Python/Numba, and Python/CuPy; Lean 4 proofs of K, F_∞, $\xi_\star$, Pillars 1–2, $\kappa = 1/6$, TheoremCLattice and `LemmaB_BetaInfinity.lean` kernel-verified in `Crossed/Transport.lean` and `Crossed/MassGap/`

($\sim$ 1900 lines, 0 `sorry`).

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